

CBET Memorization Sheet

Print this and keep it beside you. The CBET exam rewards recall of normal values, safety limits, device settings, units, standards, and formulas across every kind of equipment in a hospital. Every number on this sheet is fair game.

01 The exam itself

FACT	VALUE
Certifying body	AAMI Credentials Institute (ACI)
Questions	165 multiple choice; 150 scored, 15 unscored pretest items mixed in
Time and rules	3 hours, closed book, simple calculator allowed, no phones
Passing	Criterion-referenced cut score (modified Angoff), about 116 of 165 correct, roughly 70%
Healthcare Technology and Function	30%
Healthcare Technology Problem Solving	30%
Healthcare Information Technology	17%
Public Safety in the Healthcare Facility	10%
Anatomy and Physiology	7%
Fundamentals of Electricity and Electronics	6%

Function and problem solving are 60% of the exam. For every device: what it measures or delivers, its settings and normal values, its alarms, its common faults, and the first check.

02 Anatomy and physiology

VITAL SIGN (ADULT, RESTING)	NORMAL RANGE	MEASURED BY
Heart rate	60 to 100 bpm (below 60 bradycardia, above 100 tachycardia)	ECG, pulse oximeter, NIBP
Respiratory rate	12 to 20 breaths per minute	Impedance pneumography, capnography
Blood pressure	About 120/80 mmHg; MAP = diastolic + (systolic minus diastolic) / 3, about 70 to 100	NIBP, arterial line
Oxygen saturation (SpO2)	95 to 100%	Pulse oximeter
End-tidal CO2 (EtCO2)	35 to 45 mmHg	Capnograph
Body temperature	37 C (98.6 F); fever above about 38 C	Thermistor, thermocouple, infrared
Fetal heart rate	110 to 160 bpm	Doppler ultrasound
Tidal volume	About 500 mL at rest (6 to 8 mL/kg ideal body weight on a ventilator)	Ventilator, spirometer
Cardiac output	About 5 L/min = heart rate x stroke volume (about 70 mL)	Derived

SYSTEM	KEY PARTS	WHAT THE TECHNICIAN'S DEVICES DO WITH IT
Circulatory	Heart (right atrium, right ventricle, left atrium, left ventricle), tricuspid, pulmonary, mitral, aortic valves, arteries, veins, capillaries	ECG records electrical activity; NIBP and arterial lines measure pressure; defibrillators and pacemakers control rhythm; balloon pumps assist

SYSTEM	KEY PARTS	WHAT THE TECHNICIAN'S DEVICES DO WITH IT
Blood flow	Body, vena cava, RA, tricuspid, RV, pulmonary valve, pulmonary artery, lungs, pulmonary veins, LA, mitral, LV, aortic valve, aorta, body	Right side pumps to lungs, left side to body
Conduction path	SA node, AV node, bundle of His, bundle branches, Purkinje fibers	P wave atrial depolarization, QRS ventricular depolarization, T wave ventricular repolarization
Respiratory	Trachea, bronchi, bronchioles, alveoli, diaphragm	Ventilators move gas; oximeters measure oxygen carried; capnographs measure CO ₂ exhaled; spirometers measure volumes
Nervous	Brain (cerebrum, cerebellum, brainstem), spinal cord, peripheral and autonomic nerves	EEG records brain activity; EMG records muscle; stimulators test conduction; sympathetic raises heart rate, parasympathetic lowers it
Musculoskeletal	206 bones, joints, skeletal (voluntary), smooth (involuntary), and cardiac muscle	Physical therapy stimulators, traction, continuous passive motion
Gastrointestinal	Esophagus, stomach, small intestine (absorption), large intestine, liver, pancreas, gallbladder	Feeding pumps deliver enteral nutrition; suction removes; endoscopes view
Endocrine	Pancreas (insulin, glucagon), thyroid, adrenals (epinephrine), pituitary	Glucose meters, insulin pumps
Renal	Kidneys (nephrons filter blood), ureters, bladder	Dialysis replaces filtration
Skin	Epidermis, dermis, subcutaneous layer; barrier, temperature control, sensation	Electrode contact and skin impedance, burns, pressure injuries
Blood	Plasma, red cells (hemoglobin carries oxygen), white cells (immunity), platelets (clotting)	Analyzers count and measure; oximetry depends on hemoglobin

Directional terms and word parts

- Anterior front, posterior back; superior above, inferior below; medial toward the midline, lateral away; proximal near the trunk, distal far; supine face up, prone face down; planes: sagittal (left and right), coronal (front and back), transverse (top and bottom)
- Systole contraction, diastole relaxation. Hypertension high pressure, hypotension low. Hypoxia low oxygen, hypercapnia high CO₂, apnea no breathing
- Prefixes: brady- slow, tachy- fast, hyper- high, hypo- low, a-/an- without, dys- difficult, peri- around, endo- inside
- Roots: cardio heart, pulmo/pneumo lung, nephro/reno kidney, hepat liver, neuro nerve, gastro stomach, derm skin, osteo bone, hemo/hemat blood, myo muscle, encephal brain
- Suffixes: -itis inflammation, -ectomy removal, -ostomy opening, -oscopy viewing, -graphy recording, -gram the record, -algia pain, -emia blood condition

ECG timing (normal adult)

- PR interval 0.12 to 0.20 s; QRS under 0.12 s; QT under about 0.44 s
- Paper speed 25 mm/s; 1 mV = 10 mm; small box 0.04 s (1 mm), large box 0.20 s (5 mm)
- Heart rate = 1500 / small boxes between R waves, or 300 / large boxes
- Ventricular fibrillation: chaotic, no QRS, shockable. Asystole: flat line, not shockable. Atrial fibrillation: irregular, no P waves, cardiovert synchronized

ANATOMY DETAIL	FACTS
Skin layers	Epidermis (stratum basale at its base) over dermis over hypodermis (deepest); sebaceous glands oil, sweat glands cool, vitamin D made in skin
Muscle types	Skeletal striated voluntary; cardiac striated involuntary; smooth non-striated involuntary (organs, vessels, uterus, bladder)

ANATOMY DETAIL	FACTS
Skeleton	206 bones: 80 axial (skull, 33 vertebrae, ribs, sternum with manubrium and xiphoid, ossicles, hyoid) and 126 appendicular (scapula, clavicle, pelvis, humerus, radius, ulna, femur, patella, tibia, fibula)
Brain	Forebrain: cerebrum (frontal plans, parietal senses, occipital sees, temporal hears), thalamus relays, hypothalamus drives the pituitary. Midbrain relays vision and hearing. Hindbrain: cerebellum coordinates, pons relays, medulla runs breathing and heart rate. Brain stem = midbrain + pons + medulla
Nerves	Somatic voluntary; autonomic involuntary (sympathetic fight or flight, parasympathetic rest and digest); neurons signal by action potentials across synapses
Digestion in order	Mouth (amylase, lipase), esophagus (epiglottis, lower esophageal sphincter), stomach (pH 1.5 to 3.5, pyloric sphincter releases chyme), duodenum, jejunum, ileum, large intestine (reabsorbs water); liver makes bile, glycogen, urea; gallbladder stores bile; pancreas enzymes and insulin
Lymph and immunity	Lymph filtered in nodes (neck, armpits, groin); tonsils, thymus (T cells), spleen (cleans blood), Peyer's patches; the immune system avoids, detects, and eliminates infection
Endocrine	Hypothalamus and pituitary control; thyroid metabolism; parathyroid (posterior thyroid lobes) raises calcium; adrenal cortex cortisol, medulla epinephrine; islets insulin and glucagon; pineal rhythms; negative feedback stabilizes, positive amplifies; the pons secretes nothing
Kidney	Cortex, medulla, pelvis; nephron: glomerulus in Bowman's capsule (blood pressure drives filtration), tubules reabsorb and secrete; ureter, bladder, urethra
Reproduction	Testes, epididymis, vas deferens, prostate; FSH sperm, LH testosterone. Ovaries (estrogen, progesterone), fallopian tube (fertilization), uterus (implantation)
Blood terms	Hemoglobin carries oxygen; hematocrit is the red-cell fraction (about 45%); about 5 liters total; lymphocytes, neutrophils, monocytes, eosinophils, basophils

03 Electricity and electronics

FORMULA	STATEMENT
Ohm's law	$V = I \times R$; $I = V / R$; $R = V / I$
Power	$P = V \times I = I^2 \times R = V^2 / R$ (watts)
Series resistance	$R_{\text{total}} = R_1 + R_2 + \dots$; same current through each; voltages add
Parallel resistance	$1 / R_{\text{total}} = 1/R_1 + 1/R_2 + \dots$; two resistors: $(R_1 \times R_2) / (R_1 + R_2)$; same voltage across each; currents add
Kirchhoff	Currents into a node equal currents out; voltages around a loop sum to zero
AC values	$\text{RMS} = 0.707 \times \text{peak}$; $\text{peak} = 1.414 \times \text{RMS}$; peak-to-peak = $2 \times \text{peak}$; US mains 120 V RMS at 60 Hz (many countries 230 V at 50 Hz)
Transformer	$V_{\text{secondary}} / V_{\text{primary}} = N_{\text{secondary}} / N_{\text{primary}}$; step-down lowers voltage and raises current; isolation transformer 1:1
Capacitive reactance	$X_c = 1 / (2 \times \pi \times f \times C)$; capacitors pass AC, block DC
Inductive reactance	$X_L = 2 \times \pi \times f \times L$; inductors pass DC, oppose changes in current
RC time constant	$\tau = R \times C$; capacitor reaches 63% of full charge in one tau, about 99% in five
Frequency and period	$f = 1 / T$; 60 Hz has a period of 16.7 ms
Stored energy	$E = \text{one half} \times C \times V^2$ (joules); how a defibrillator capacitor stores its charge
Energy	$E = P \times t$; 60 W for 8 h = 0.48 kWh; kWh is energy, kW is power

FORMULA	STATEMENT
Voltage divider	V across R1 = V source x R1 / R total (series); 120 V across 60 and 100 ohms puts 45 V on the 60
Current divider	I through a branch = I total x (1/R branch) / (sum of 1/R); the smallest resistance takes the most current
Kirchhoff's laws	KCL: current into a node equals current out (nodal analysis). KVL: voltages around a loop sum to zero (mesh analysis)
Internal resistance	(open-circuit voltage minus loaded voltage) / load current; 1.58 V to 1.46 V into 80 ohms = 6.58 ohms; rises with age
Wheatstone bridge	Balanced (no meter current) when R1 / R2 = R3 / R4; strain gauges and thermistors sit in bridges
Complex impedance	Z = R + jX; 40 mH at 60 Hz is about 15 ohms of reactance; an ideal 30 mH inductor at 60 Hz is 11.3j ohms
AC power	S squared = P squared + Q squared; P true (W), Q reactive (VAR), S apparent (VA); 180 W and 240 VAR give 300 VA; power factor = P / S
Op-amp gain	Inverting gain = R feedback / R input (gain 9 with 5 k in needs 45 k feedback); non-inverting = 1 + Rf / Rin
Transformer ratio	V primary / V secondary = N primary / N secondary; 240 V to 6,000 V is a 0.04 ratio
C-rate	Charge or discharge current relative to capacity; 60 kWh at 0.5 C = 30 kW

- Resistor color code digits: black 0, brown 1, red 2, orange 3, yellow 4, green 5, blue 6, violet 7, gray 8, white 9; tolerance gold 5%, silver 10%
- US cord colors: black (or red) hot, white neutral, green or bare ground. IEC cords: brown live, blue neutral, green-yellow ground
- Schematic symbols: zigzag resistor, parallel lines capacitor, coil inductor, triangle-with-bar diode, ground, fuse, switch, battery; signals flow left to right; compare measured voltages to the values printed at test points
- Passive devices (resistors, capacitors, inductors, transformers) cannot add energy; active devices (diodes, transistors, ICs, op-amps) control or amplify
- Diode conducts one way, forward drop about 0.7 V silicon; zener regulates; LED lights; rectifiers: half-wave, full-wave, bridge; capacitor filters ripple; regulator holds output steady
- Linear power supply: heavy transformer, low noise, inefficient. Switching supply: light, efficient, more noise, needs a load
- Transistor: NPN or PNP, base controls collector-emitter; used as a switch or amplifier. SCR and triac switch AC power
- Op-amp rules: huge gain, inputs draw no current, feedback makes the inputs equal; instrumentation amplifier rejects common-mode noise (high CMRR) for ECG front ends; isolation amplifiers and optocouplers keep patient circuits floating
- Filters: low-pass keeps slow signals, high-pass removes baseline drift, band-pass keeps a range, notch removes 60 Hz
- DMM measures volts, amps (in series), ohms (power off); oscilloscope shows voltage against time, 10x probe divides by ten; function generator provides test signals
- Passive: resistor, capacitor, inductor, memristor, thermistor (no gain). Active: transistor, op-amp, IC, SCR. Transducer converts energy: sensor to electrical, actuator from electrical; piezoelectric crystals do both
- US distribution: step-up to transmit, step-down to use; split-phase 120/240 V at 60 Hz with a grounded center neutral; NEC is NFPA 70 (building wiring), NESC is IEEE (utility lines)
- Battery specifications: voltage per cell, capacity (Ah), energy density (sets weight), internal resistance (rises with age, sag under load), shelf life, discharge curve, cycle life, C-rate
- Regulators: linear drops excess as heat (quiet, inefficient); switching chops at a duty cycle (efficient, noisy); shorted switching element passes full input voltage, open element gives zero; test regulator input and output pins to ground
- Filters by what they pass: low-pass, high-pass, band-pass, band-stop (notch is a narrow band-stop)

POWER STORAGE AND CONDITIONING	FACTS
Cell voltages	Lead-acid 2.0 V, nickel-cadmium 1.2 V, nickel-metal hydride 1.2 V, lithium-ion 3.6 to 3.7 V, alkaline 1.5 V
Battery behavior	NiCd memory effect (condition with full cycles); Li-ion no memory but thermal runaway risk and must not over-discharge; sealed lead-acid sulfates if left discharged; capacity in amp-hours falls with age
UPS types	Standby (switches on loss), line-interactive (regulates voltage), online double-conversion (always on inverter, best protection)
Power problems	Sag or brownout (low voltage), surge (high), transient spike, noise, blackout; conditioners regulate and filter

POWER STORAGE AND CONDITIONING	FACTS
Hospital essential electrical system (NFPA 99 and NFPA 70)	Generator restores power within 10 seconds; branches: life safety, critical (patient care, red outlets), equipment; UPS bridges the gap for critical devices
Isolated power and LIM	Ungrounded secondary of an isolation transformer; line isolation monitor alarms at 5 mA total hazard current; used in wet procedure locations
GFCI	Trips at 4 to 6 mA of imbalance between hot and neutral in about 25 ms; wet locations

04 Electrical safety

CURRENT THROUGH THE BODY (60 HZ, HAND TO HAND)	EFFECT
1 mA	Threshold of perception (tingle)
5 mA	Accepted maximum harmless current
10 to 20 mA	Let-go threshold: muscles contract and the person cannot release
50 mA	Pain, possible fainting, respiratory arrest
100 to 300 mA	Ventricular fibrillation likely
6 A and above	Sustained heart contraction, burns, respiratory paralysis
Microshock	As little as 10 microamps delivered directly to the heart (through a catheter or pacing wire) can cause fibrillation

IEC 60601-1 LIMIT	NORMAL CONDITION	SINGLE FAULT CONDITION
Earth (ground) leakage	5 mA	10 mA
Touch (enclosure, chassis) current	100 microamps	500 microamps
Patient leakage, type B and BF, AC	100 microamps	500 microamps
Patient leakage, type CF, AC	10 microamps	50 microamps
Patient leakage, DC (all types)	10 microamps	50 microamps
Mains voltage on the applied part	BF: 5 mA; CF: 50 microamps	(single fault test)

NFPA 99 AND FACILITY FACTS	VALUE
Ground (chassis to plug ground pin) resistance	0.5 ohm maximum for patient-care equipment
Chassis (touch) leakage, cord-connected patient-care equipment	500 microamps in current editions (older editions used 300 or 100 microamps; know the limit your facility cites)
Patient care vicinity	About 6 feet (1.8 m) beyond the bed or chair and 7.5 feet (2.3 m) above the floor
Line isolation monitor alarm	5 mA total hazard current
GFCI trip	4 to 6 mA (nominal 5 mA)
Hospital-grade receptacle	Green dot; test retention force, polarity, ground continuity
Relocatable power taps	Only where allowed, rated, and secured; never daisy-chained; no extension cords for patient care equipment
Relocatable power tap load	No more than 75% of the tap's rating (an AWG 10 tap rated 35 A carries at most 26 A); never daisy-chained

NFPA 99 AND FACILITY FACTS	VALUE
Prohibited	Three-to-two prong (cheater) adapters; tinned stranded wire ends under screw terminals; cords without strain relief; isolated-ground receptacles in the patient care vicinity
Patient room receptacles	At least 8 per general bed (more in critical care), non-locking, tamper-resistant; no minimum in bathrooms; OR receptacles single, duplex, or quadruplex with insulated grounds, locking or not
Household electronics	Allowed only outside the patient care vicinity unless double insulated; inspected before patient use; worn units removed
Gas cylinder storage	Unwrapped, secured upright, minus 7 to 52 C; oxidizers 20 ft from flammables (5 ft with sprinklers, none in a compliant cabinet); no smoking within 20 ft; not in tight enclosures; door warning for other positive-pressure gases
Medical gas systems	DOT/TC/ASME cylinders; gas-specific outlets, no adapters; redundant supply; alarms; labeled zone shutoff valves; inventory and inspection
Telecommunications rooms	No data outlet more than 90 m from its room; at least one per floor; each serves at most 20,000 sq ft; interior rooms in disaster areas
Nurse call	Carries device alarms, staff emergencies, code calls, patient requests; audible and visual annunciation; may forward to pagers
Touch current and clean agents	Touch current: chassis leakage reachable by a person via a path other than ground. Clean agents: nonconductive, residue-free extinguishants (nitrogen, CO ₂ , noble gases, halocarbons)

Classes, types, symbols, and testing

- Class I: protective earth (three-prong ground). Class II: double insulation, no ground (square-in-square symbol). Internally powered: battery only
- Applied part type B: body, not for direct cardiac use (stick figure). BF: body floating, isolated (figure in a box). CF: cardiac floating, safe for direct heart connection (heart in a box)
- Defibrillation-proof applied part: the symbol with paddles; withstands a defibrillator discharge
- Macroshock: current through the skin; needs milliamps. Microshock: current bypassing the skin to the heart; microamps
- Safety analyzer sequence: visual inspection and cord check, ground resistance, chassis leakage (normal, reversed polarity, open ground, open neutral), patient lead leakage, lead-to-lead, mains on applied part
- Leakage rises with open ground and reversed polarity; a rising leakage trend on PM predicts insulation failure; test at incoming inspection, after repair, and per PM schedule

05 Infection control, hazards, fire, gas, radiation, MRI, and standards

Infection control

- Standard (universal) precautions: treat all blood and body fluids as infectious; hand hygiene before and after every patient contact and before touching equipment
- Transmission-based precautions: contact (gown, gloves), droplet (surgical mask), airborne (N95 respirator, negative-pressure room)
- PPE donning: gown, mask, eye protection, gloves. Doffing: gloves, eye protection, gown, mask (clean hands between)
- Cleaning removes soil; disinfection kills most microbes; sterilization kills all including spores
- Spaulding: critical items (enter sterile tissue) sterilize; semi-critical (touch mucous membranes) high-level disinfect; non-critical (intact skin) low-level disinfect
- Steam autoclave: 121 C (250 F) at 15 psi for 15 to 30 minutes, or 132 to 135 C flash cycles; biological indicators verify; ethylene oxide and hydrogen peroxide plasma for heat-sensitive items
- Clean and disinfect equipment before servicing it; sharps in puncture-resistant containers; OSHA bloodborne pathogens standard covers exposure control; report needlesticks immediately
- Universal precautions (1980s): blood and blood-contaminated fluids only, not sweat, tears, nasal secretions, or urine; bleach cleanup; bagged waste. Standard precautions (CDC 1996): all body fluids plus body substance isolation; transmission-based precautions supplement them

- Regulated waste: liquid or free-flowing blood, sealed for a licensed medical waste service; dried blood on dressings sealed in a plastic bag with regular trash; broken glass with tools, never hands
- Lab practice: annual biological risk assessment sets the biosafety level; no mouth pipetting; no food, drink, or contact lenses; retractable needles; decontaminate before disposal; eyewash station; hands-free sinks; clean utility room separate from dirty utility; lab coats are not gowns

Hazard communication and signage

- Safety data sheet (SDS): 16 sections including identification, hazards, first aid, fire fighting, handling and storage, exposure controls and PPE; must be accessible to staff
- GHS pictograms: flame (flammable), flame over circle (oxidizer), skull (acute toxicity), corrosion, exclamation mark (irritant), health hazard (silhouette), gas cylinder, environment
- NFPA 704 diamond: blue health, red flammability, yellow reactivity, white special (W with a line means water reactive, OX oxidizer); scale 0 to 4
- Laser signs state the class and wavelength; classes 1 (safe) through 4 (fire and eye hazard); wavelength-specific eyewear
- Ionizing radiation: trefoil symbol; time, distance, shielding; ALARA; inverse square law; lead aprons about 0.5 mm; dosimeters
- MRI: zones I (public) through IV (magnet room); the magnet is always on; ferromagnetic objects become projectiles; quench vents helium; MR Safe, MR Conditional, MR Unsafe labels
- Patient precaution signs at the door: contact, droplet, airborne, fall risk, latex allergy, no blood pressure on a limb
- HazCom (OSHA 29 CFR 1910.1200) requires labels, SDS access, training, and records; it adopted the UN Globally Harmonized System (GHS) for classification, pictograms, signal words, and the 16-section SDS
- SDS order: identification, hazards, composition, first aid, fire fighting, accidental release, handling and storage, exposure controls and PPE, physical and chemical properties, stability and reactivity, toxicology, ecology, disposal, transport, regulatory, other
- Toxicity and limits: LD50 lethal dose for 50%, LC50 lethal concentration; PEL legal limit (OSHA); TLV recommended (ACGIH); REL recommended (NIOSH); STEL 15-minute ceiling; carcinogen lists from NTP, IARC, OSHA
- GHS label: identity, manufacturer contact, pictograms, signal word (Danger or Warning only), hazard and precautionary statements; secondary containers need the same; pyrophoric ignites in air within 5 minutes; oxidizer pictogram exists, reducer does not

Fire and medical gas

- Fire triangle: heat, fuel, oxygen. RACE: rescue, alarm, confine, extinguish. PASS: pull, aim, squeeze, sweep
- Extinguisher classes: A ordinary combustibles, B flammable liquids, C electrical, D metals, K cooking oils
- Oxygen-enriched atmospheres burn faster; no oil or grease on oxygen fittings; cylinders secured upright, capped, stored below 125 F, full separated from empty
- US medical gas cylinder colors: oxygen green, nitrous oxide blue, carbon dioxide gray, nitrogen black, medical air yellow, helium brown; pin index and DISS fittings prevent wrong connections
- Full E cylinder of oxygen: about 2,000 to 2,200 psi, about 660 liters; wall outlets about 50 psi; zone valves shut off gas to an area

STANDARDS, REGULATORS, AND ACCREDITATION	WHAT TO KNOW
CMS	Centers for Medicare and Medicaid Services; sets Conditions of Participation hospitals must meet to be paid; can survey directly
Deemed status	Accrediting organizations survey on CMS's behalf: The Joint Commission, DNV Healthcare, ACHC (formerly HFAP), CIHQ
The Joint Commission	Environment of Care and Emergency Management standards cover equipment inventory, maintenance strategies, testing, and documentation
FDA	Regulates devices; medical device reporting: user facilities report a device-related death to FDA and the manufacturer within 10 work days, a serious injury to the manufacturer within 10 work days; recalls Class I (serious harm), II (temporary harm), III (unlikely harm)
OSHA	Worker safety: hazard communication, bloodborne pathogens, PPE, electrical safety in the workplace
ANSI/AAMI EQ56	Recommended practice for a medical equipment management program
ANSI/AAMI EQ89	Guidance for scheduled maintenance and performance verification

STANDARDS, REGULATORS, AND ACCREDITATION	WHAT TO KNOW
ANSI/AAMI EQ93	Medical equipment management vocabulary and terminology
ANSI/AAMI EQ103	Alternative equipment maintenance (AEM) programs: how to justify and document maintenance strategies that differ from the manufacturer's
AEM rules (CMS)	Allowed with documented risk assessment by qualified staff; not for imaging or radiologic equipment, lasers, or where the manufacturer or law requires otherwise; new equipment follows the manufacturer until history supports a change
CMMS	Computerized maintenance management system: inventory and asset tags, work orders, PM schedules, equipment history, parts, reports (PM completion rate, mean time between failures, cost); every activity documented there
After an incident	Remove the device with all accessories and disposables attached, do not clear logs or settings, tag and sequester, document, report; root cause analysis asks why repeatedly
Joint Commission programs	Accredits and certifies by peer survey; publishes National Patient Safety Goals yearly; links quality and risk management; does not set financial benefits
DNV	Accredits hospitals under NIAHO, integrating CMS Conditions of Participation with ISO 9001; annual surveys
ISO 9000 / 9001	Quality management standards; plan-do-check-act; risk-based thinking; customer focus, leadership, process approach
CMS programs	Medicare, Medicaid, CHIP; regulates human laboratory testing; publishes yearly core measure sets (clinical quality measures); fights fraud; does not accredit
CAP	College of American Pathologists: nonprofit accreditor of clinical laboratories under CMS authority; inspections every two years plus unannounced
SMDA	Safe Medical Devices Act (1990, amended 1992): user facilities report device-related deaths to FDA and manufacturer, serious injuries to the manufacturer; annual summaries

06 Physiological monitoring

ECG FACT	VALUE
Einthoven's triangle	Lead I = LA minus RA; Lead II = LL minus RA; Lead III = LL minus LA; Einthoven's law: Lead II = Lead I + Lead III
Augmented and chest leads	aVR, aVL, aVF from the limb electrodes; V1 to V6 across the chest; 12 leads from 10 electrodes
Electrode colors (US, AHA)	RA white, LA black, RL green (ground), LL red, V brown ("white on right, smoke over fire")
Bandwidth	Monitoring mode 0.5 to 40 Hz; diagnostic mode 0.05 to 150 Hz
Interference	60 Hz: bad electrode contact, unshielded leads, nearby equipment; wandering baseline: respiration or motion; spikes: muscle tremor or loose lead
Lead-off	High impedance detected on an electrode; check gel, skin prep, cable
Simulator test	Patient simulator provides rate, amplitude, and arrhythmias to verify the monitor
Rhythms on the strip	Sinus tachycardia fast regular; atrial (supraventricular) tachycardia rapid consistent narrow complexes; atrial fibrillation irregular with quivering waves, no P; ventricular tachycardia rapid WIDE QRS; ventricular fibrillation chaotic; asystole flat (check leads)

ECG FACT	VALUE
Lead views and counts	aVF views the inferior wall; 12 leads from 10 electrodes (4 limb, 6 chest); IEC colors RA red, LA yellow, LL green, RL black differ from AHA; thermal print head tested by printing letters or diagonals; paper speeds 10, 12.5, 25, 50 mm/s test the drive

BLOOD PRESSURE FACT	VALUE
Korotkoff sounds	Auscultatory method: cuff above systolic, deflate; first sound (phase 1) is systolic, sounds muffle then disappear (phase 5) at diastolic; automated devices use oscillometry instead
NIBP method	Oscillometric: cuff inflates above systolic, deflates in steps, MAP is the point of maximum oscillation, systolic and diastolic derived
Cuff sizing	Bladder width about 40% of arm circumference, length about 80%; too small reads high, too large reads low
Invasive line	Transducer zeroed to atmosphere and leveled at the phlebostatic axis (fourth intercostal space, midaxillary line)
Dicrotic notch	The small dip on the downslope of the arterial waveform marking aortic valve closure; its absence suggests overdamping
Height error	13.6 cm of water equals 10 mmHg; every 13.6 cm the transducer is off level changes the reading about 10 mmHg
Damping	Fast-flush (square wave) test: overdamped rounds the wave and reads low systolic; underdamped overshoots and reads high
Transducer sensitivity	5 microvolts per volt per mmHg (industry standard)
Sphygmomanometer	Cuff, bulb, valve, manometer (mercury, aneroid, electronic), stethoscope; gauge not at zero: zeroing screw; unstable: movement or placement; reads high: cuff below the heart; reads low: cuff above the heart or an air leak (most common); invasive lines read continuously even when hypotensive

OXIMETRY AND CAPNOGRAPHY	VALUE
Pulse oximeter wavelengths	Red about 650 to 660 nm, infrared about 805 to 940 nm; oxyhemoglobin absorbs more infrared; ratio of pulsatile absorbances gives SpO ₂ ; high perfusion helps, low perfusion hurts
Errors	Carboxyhemoglobin reads falsely high; methemoglobin trends toward 85%; motion, low perfusion, nail polish, ambient light, dyes
Testing	SpO ₂ simulator or optical tester; check probe LED and detector, cable, and site
Capnography	Infrared absorption by CO ₂ (about 4.26 micrometers); mainstream sensor at the airway, sidestream samples through tubing (delay, water trap)
Capnogram	Baseline near zero, rapid rise, alveolar plateau, EtCO ₂ at end of plateau, drop on inspiration; loss of waveform means disconnect, apnea, or esophageal intubation

Other monitoring

- Temperature: thermistor (resistance falls as temperature rises, NTC), thermocouple (voltage from two metals), infrared tympanic; YSI 400 and 700 series probe curves are not interchangeable
- Respiration on a monitor: impedance pneumography through the ECG electrodes; capnography is more reliable
- Alarm priorities (IEC 60601-1-8): high red flashing, medium yellow flashing, low cyan or yellow steady; alarm fatigue is a patient safety problem; alarm management programs set limits per unit
- Telemetry: WMTS bands 608 to 614 MHz, 1395 to 1400 MHz, 1427 to 1432 MHz; dropouts from antenna coverage, low battery, interference; central station shows all beds
- Fetal monitor: ultrasound Doppler for fetal heart rate (110 to 160 bpm), tocodynamometer (TOCO) for contractions by pressure on the abdomen; internal scalp electrode and intrauterine pressure catheter are the direct methods
- Respiration methods: chest impedance through ECG electrodes (most common), respiratory belts, airflow sampling (thermistor or capnograph), magnetometry; a fingertip oximeter that shuts off mid-use has a low battery
- EEG 10-20 landmarks: nasion (bridge of the nose) to inion (back of the skull); beta above 13 Hz awake and active; differential amplifier removes common signals

- EEG bands: delta under 4 Hz, theta 4 to 8, alpha 8 to 13, beta above 13; electrodes placed by the 10-20 system; EMG measures muscle electrical activity; evoked potentials time the response to a stimulus

07 Test equipment and units of measure

TEST EQUIPMENT	VERIFIES
Electrical safety analyzer	Ground resistance, chassis and patient lead leakage under normal and fault conditions
Patient (physiological) simulator	ECG rate, amplitude, arrhythmias, respiration, invasive pressure, temperature
SpO2 simulator	Saturation and pulse readings through the probe
NIBP simulator	Cuff pressures and oscillometric readings; leak test
Defibrillator analyzer	Delivered energy into 50 ohms, charge time, synchronization, pacing output
ESU analyzer	Output power into a load, leakage, return electrode monitoring
Gas flow and ventilator analyzer	Volume, flow, pressure, oxygen concentration
Infusion pump analyzer	Flow rate accuracy, occlusion pressure
DVM or DMM and meters	Voltage, current, resistance; oxygen analyzer, pressure meter, tachometer, light meter, sound level meter
Cable tracer or toner	Finds the far end of a network cable
What each analyzer does not do	Ventilator analyzer: flow, pressure, volume, O2, breath detection, no CO2 concentration. ESU analyzer: RMS power, crest factor, peak-to-peak voltage, RF leakage, REM thresholds, no handpiece continuity or chassis leakage (multimeter and safety analyzer). Defibrillator analyzer: energy, charge time, sync, not internal impedance. Simulator: no current. DVM: voltage only, draws a little current; DMM continuity = low resistance, unpowered
UNIT	CONVERSION OR MEANING
mmHg	Blood and gas pressure; 760 mmHg = 1 atmosphere; 1 mmHg = 1.36 cm H2O
cm H2O	Ventilator and airway pressures; 1 cm H2O = 0.74 mmHg
psi	Gas cylinders and pipelines; 14.7 psi = 1 atmosphere; 1 psi = 51.7 mmHg; 1 psi = 6.9 kPa
bar	1 bar = 100 kPa = 14.5 psi = 750 mmHg, about 1 atmosphere
kPa	1 kPa = 7.5 mmHg; 101.3 kPa = 1 atmosphere
Fahrenheit and Celsius	$F = C \times 1.8 + 32$; $C = (F \text{ minus } 32) / 1.8$; 37 C = 98.6 F; 0 C = 32 F; 100 C = 212 F
Joules	Energy: 1 J = 1 watt-second; defibrillator output
Lumens and lux	Lumens: total light output of a lamp; lux: lumens per square meter at the surface (surgical lights are rated in lux at 1 m)
Decibels	Sound level and audiometer hearing threshold (dB HL); 10 dB is a tenfold power ratio
Weight and length	1 kg = 2.2 lb; 1 inch = 2.54 cm; 1 L = 1000 mL; 1 mL = 1 cc

UNIT	CONVERSION OR MEANING
Prefixes	kilo 10 to the 3, milli 10 to the minus 3, micro 10 to the minus 6, nano 10 to the minus 9

08 Diagnostic, infusion, therapeutic, and laboratory equipment

DIAGNOSTIC DEVICE	FACTS
Otoscope, ophthalmoscope	Battery or wall-powered light and optics for ear and eye; check lamp, battery, charging contacts, specula
Audiometer	Hearing thresholds in dB HL at 250 to 8000 Hz; calibrate with a sound level meter and coupler; quiet room or booth
Spirometer	FVC, FEV1, FEV1/FVC ratio (normal about 0.7 to 0.8); calibrate with a 3 L syringe; filters and mouthpieces
Stress test system	Treadmill or bike with 12-lead ECG and NIBP; emergency stop; treadmill speed and grade calibration; defibrillator nearby
Scales	Calibrate with certified test weights; zero with the bed or chair empty; load cells
Ultrasound	Piezoelectric transducer sends and receives; 2 to 15 MHz; higher frequency, better resolution, less penetration; Doppler measures flow; gel removes air; transducer crystals crack if dropped
Otoscope parts	Handle, head, lamp with collar, magnifying lens, specula, insufflation port; no microphone; loose head cuts light off, worn fibers dim it
Ophthalmoscope	Views the fundus; lens disc, filters, mirror, condensing lens; clean with lens wipes, alcohol discolors
Audiogram	Threshold in dB versus frequency; air conduction (headphones) versus bone conduction (mastoid vibrator) separates sensorineural from conductive loss; creates tones, does not measure frequencies
Uroflow meter	Flow rate and volume of voiding (bladder emptying); funnel, beaker, transducer; wireless within about 3 m
Tympanic thermometer	Infrared from the eardrum onto a thermopile; probe covers; 30 minutes to settle; ambient 10 to 40 C; torn cover reads high, cracked lens or obstruction reads low
Endoscope	Fiber-optic or camera, cool light source, insufflation, irrigation, suction; leak test before immersion; no wireless in flexible scopes
Imaging	X-ray: electrons from a tungsten filament strike a tungsten anode; radiopaque bone is white. Fluoroscopy: live X-ray through an image intensifier. CT: rotating tube and detectors, computed slices. MRI: magnet aligns hydrogen, RF, realignment signal, no ionizing radiation, pacemakers contraindicated. PET: FDG tracer, positron annihilation, two gamma photons. Ultrasound Doppler: toward the probe = higher pitch

INFUSION DEVICE	FACTS
Large-volume pump	Peristaltic; accuracy typically within 5%; rate in mL/hr; free-flow protection; occlusion and air-in-line alarms; drug library
Syringe pump	Precise small volumes; syringe size must be recognized; start-up delay and siphoning risks
PCA pump	Patient button, bolus dose, lockout interval, hourly limit; locked to prevent tampering
Feeding (enteral) pump	Enteral-only connectors on purpose; lower accuracy acceptable
Contrast injector	Programmed volume and flow for imaging; pressure limit; air detection; heated syringe
Rate math	$\text{mL/hr} = \text{total mL} / \text{hours}$; $\text{drops per minute} = (\text{mL/hr} \times \text{drop factor}) / 60$; $\text{mL/hr} = (\text{mg/kg/hr} \times \text{kg}) / (\text{mg/mL})$
Enteral versus parenteral	TPN by vein through a standard pump; enteral by rotary peristaltic head on wide tubing; low flow = empty bag, occlusion, drip chamber seated wrong or coated; no set = seating, wrong set, or bright light on the detector

INFUSION DEVICE	FACTS
Syringe pump settings	Must know the syringe diameter and brand; small volume, fine control; runs past empty = sensor, size setting, or multi-phase program, not low voltage
PCA settings	Bolus, lockout interval, maximum dose per hour or 4 hours, optional background; logged; locked; ignores requests = check settings first
Infusion modes	Continuous; intermittent (high-rate periods separated by low-rate intervals); bolus; secondary or piggyback; pressure infusers squeeze a bag and are not pumps
THERAPEUTIC DEVICE	FACTS
Infant incubator and warmer	Air and skin temperature servo control, humidity, oxygen; radiant warmer with skin probe; over-temperature alarms; phototherapy blue light about 425 to 475 nm
Patient temperature management	Hypo- and hyperthermia units circulate water through blankets or pads; water temperature limits about 4 to 42 C; check flow, leaks, temperature sensor
Aspiration (suction)	Regulators in mmHg, continuous or intermittent; adult range about 80 to 120 mmHg; check filter, collection canister, tubing, vacuum source
Sequential compression device	Inflates leg sleeves in sequence to prevent clots; check pressures and cycle timing, hoses, sleeves
Physical therapy equipment	TENS and muscle stimulators, therapeutic ultrasound (1 to 3 MHz), diathermy, traction, continuous passive motion; output and timer verification
Radiant warmer versus incubator	Radiant: open bed, overhead heater, skin sensor servo, easy access, drafts and dehydration. Incubator: enclosed, air, humidity, oxygen control. Fluctuating temperature = sensor attachment
Ultrasound therapy	1 to 3 MHz deep heat for pain and blood flow; needs gel and motion; head too cold is not detected; overheating shutdown = too little gel; slow warming = low drive voltage
Hypo/hyperthermia flow	Heating needs flow of about 1 L/min or more, correct limits, and a good element; poor heating and cooling with no leaks = insufficient flow
Aspirator parts	Vacuum pump, gauge or regulator, tubing, collection container with overflow float, hydrophobic bacterial filter; pulmonary, gastric, surgical; fading suction = leak or filter; sudden stop = fuse or overheated pump
SCD and bilirubin	SCD inflates distal (ankle) to proximal (thigh), holds seconds, rests about a minute; high pressure = sleeve too tight. Bilirubin light 360 to 520 nm, blue-green most effective, brighter, closer, more skin; bulb, ballast, fan filter
LABORATORY DEVICE	FACTS
Centrifuge	Speed in RPM, force in RCF (g); balance opposite loads; lid interlock; verify speed with a tachometer and timer
Incubator	Temperature (often 37 C) and sometimes CO ₂ and humidity; verify with a reference thermometer; door seals
Rocker and shaker	Speed and tilt; motor and belt
Refrigerator and freezer	Blood and vaccine storage 2 to 8 C; freezers minus 20 or minus 80 C; continuous temperature logging and alarms; door gaskets and defrost
Microscope	Light source, objectives, focus; cleaning optics
Water bath	Temperature control and uniformity; thermostat and heater
Analyzers	Reagents, calibration, quality control samples, fluidics and clogs, temperature
Cryostat and microtome	Cryostat cuts frozen sections in a cold chamber (about minus 20 C); microtome cuts thin sections; blade safety, temperature, advance mechanism
Blood gas analyzer	Clark electrode pO ₂ (platinum cathode, silver anode, O ₂ membrane, current); glass electrode pH (voltage); Severinghaus electrode pCO ₂ (pH electrode behind a CO ₂ membrane in bicarbonate); too cold = thermistor; barcode = window

LABORATORY DEVICE	FACTS
Spectrophotometer	Absorbance at selected wavelengths; tungsten lamp for visible, deuterium for UV; read a blank first; quartz cuvettes for UV; intermittent at all wavelengths = power cable
Hematocrit centrifuge	Packs red cells in capillary tubes at about 10,000 to 15,000 g; RCF rises with speed squared and radius
Microtome detail	Paraffin-embedded blocks; thickness depends on blade clamping, sharpness, clearance angle, cool block, slow steady stroke, no wax; cryostat frost = drafts or breath, splinter = too cold
Water bath	Gentle heating under 100 C with stirring; incubates microbial samples; not heating = setpoint or limit below ambient, or low water

09 Perioperative and life-support devices

ELECTROSURGERY FACT	VALUE
Frequency	About 300 kHz to 3 MHz, above the frequency that stimulates nerves and muscle
Modes	Cut: continuous sine wave, vaporizes. Coag: interrupted or damped bursts, dries and seals. Blend mixes them
Monopolar	Current flows from the active electrode through the patient to a large return (dispersive) electrode; return electrode contact quality monitoring prevents burns
Bipolar	Current flows between the two tips of the forceps; no return pad needed
Burns	Poor return pad contact, pad on bony or hairy site, alternate paths through ECG electrodes or metal; capacitive coupling in laparoscopy
Video integration	Cameras, light sources, monitors, recording and routing systems; check cable, connector, light cable fibers, white balance
Waveform and faults	Cut = continuous sine, 100% duty cycle, monopolar; coag = pulsed or damped, low duty cycle, higher peak; on but no output = blown high-voltage fuse; will not turn on = cord, fuse, low-voltage supply, switch; footswitch dead = footswitch board then control board

Other perioperative equipment

- Pneumatic tourniquet: cuff pressure above systolic (commonly 250 mmHg arm, 300 mmHg leg or adjusted to the patient), time limit alarms (about 2 hours), leak test, calibrated gauge
- Fluid and blood warmer: keeps fluids near body temperature; over-temperature cutoff around 41 to 42 C; check heater, sensor, alarms
- Sterilizer: steam 121 C at 15 psi or 132 C flash; verify temperature, pressure, time, gasket, chamber; biological indicator; ethylene oxide and hydrogen peroxide for heat-sensitive items
- OR table: hydraulics or motors, controls, brakes, weight rating; surgical lights: lux at 1 m, color temperature, handles, bulbs or LEDs; surgical microscope: optics, illumination, balance, focus motors
- Sterilization methods: steam (fastest; dry heat for items that cannot be wet), ethylene oxide gas (heat- and moisture-sensitive, about 10 h plus aeration), liquid chemical (small hard-to-reach lumens, rinse), ionizing gamma or beta (disposables in sealed packages); no steam = water pump, low fluid pressure, breaker, heater
- Surgical lasers: CO2 far infrared absorbed by water for skin and soft tissue; Nd:YAG near infrared penetrates (hair, tattoo, coagulation); argon blue-green for pigment and eye; excimer UV for corneas; aiming beam and articulated arm; aiming only = footswitch
- Blood warmer: countercurrent warm water around the blood line; pressurized-flow units add a compressor (loud = unseated tubing or failing compressor; hot = blocked vents); prevents transfusion hypothermia, not for lab samples or infants

- Heart-lung machine: venous reservoir, roller or centrifugal pump (gentle on cells), membrane oxygenator (older bubble), heat exchanger for hypothermia, filters, arterial return; heparin thins blood; cardioplegia stops the heart; backup pumps
- Hemodialysis order: access, arterial monitor, blood pump, heparin pump, dialyzer (countercurrent dialysate across a semipermeable membrane), venous monitor, air trap and detector (last), return

DEFIBRILLATION AND PACING	VALUE
Adult energy	Monophasic 360 J; biphasic 120 to 200 J (device specific); pediatric 2 J/kg first, 4 J/kg after
Synchronized cardioversion	Shock timed to the R wave for organized rhythms (atrial fibrillation, SVT); unsynchronized for VF and pulseless VT
Testing	Defibrillator analyzer with a 50 ohm load; check delivered energy, charge time, synchronization, pacing output; daily user checks
AED	Analyzes rhythm, advises shock only for VF and pulseless VT
External pacing	Demand mode paces only when intrinsic rate falls below the set rate; fixed mode paces regardless; output in mA; capture confirmed by QRS after each spike
Intra-aortic balloon pump	Balloon in the descending aorta inflates in diastole (raises coronary perfusion) and deflates just before systole (lowers afterload); timed to the ECG or arterial waveform; helium gas; alarms for timing, leak, and gas loss
Pacing routes	Transcutaneous: chest pads with gel, tens of mA, painful, fastest. Transvenous: lead through a vein (subclavian) into the right ventricle, low output. Epicardial: wires on the heart surface at surgery, no subclavian leads. External generators run on AC or battery

VENTILATOR AND ANESTHESIA	VALUE
Tidal volume and minute volume	6 to 8 mL/kg ideal body weight; minute volume = tidal volume x rate
Typical settings	Rate 12 to 20; I:E ratio 1:2; PEEP 5 cm H ₂ O; FiO ₂ 0.21 to 1.0
Modes	Volume control delivers a set volume; pressure control a set pressure; assist-control, SIMV, pressure support, CPAP; BiPAP has separate inspiratory and expiratory pressures
Alarms	High pressure: obstruction, kink, coughing, secretions. Low pressure or low volume: disconnect or leak. Apnea: no breath detected
Oxygen analyzer calibration	Room air 21% and 100% oxygen; galvanic cells wear out; concentrators deliver about 90 to 95%
Anesthesia machine	Pipeline (50 psi) and cylinder supply, flowmeters, agent-specific vaporizers (sevoflurane yellow, isoflurane purple, desflurane blue), hypoxic guard keeps oxygen at least 21 to 25%, oxygen flush, CO ₂ absorber, scavenging removes waste gas, pin index and DISS prevent misconnection; leak test before use
Anesthesia gas path	Supply (pipeline 50 psi, cylinders), regulators, flow controls and flowmeters, vaporizer (first section after the agent's flow control), common gas outlet, breathing circuit with CO ₂ absorber, ventilator, patient; scavenging removes waste

10 Problem solving and troubleshooting

The method, in order

- 1. Confirm the complaint: talk to the user, reproduce the problem, rule out use error (wrong settings, wrong accessory, wrong technique), and check the right disposables.
- 2. Check the simple things: power, plug, battery, fuse, cables, connectors, settings, sensors.
- 3. Isolate: which subsystem (power, sensor, signal path, processor, display, mechanical); half-split the path with a simulator and test equipment; localized device or system-wide?

- 4. Form and test one hypothesis at a time; swap known-good parts; use the schematic.
- 5. Repair or replace; follow the service manual.
- 6. Verify: full performance test and electrical safety test after any repair, before return to service.
- 7. Document in the CMMS: symptom, cause, action, parts, test results, time. Educate the user if it was a use error.

SYMPTOM	FIRST SUSPECTS
Device dead, no power	Outlet, cord, plug, fuse, power switch, battery, power supply output
Runs on AC but not battery	Battery aged or not charging; charger circuit; battery connector
Intermittent reboot or errors	Failing power supply capacitors, loose connector, overheating
ECG shows 60 Hz noise	Electrode contact, gel, skin prep, lead cable shield, nearby equipment, ground
Wandering baseline	Patient movement or respiration, loose electrode, high-pass filter setting
Intermittent signal	Cable flex failure near connectors, loose connector, cracked solder
NIBP reads high	Cuff too small, cuff too low, patient movement, hose leak causing slow deflation
Arterial line reads low and rounded	Overdamped: air bubble, clot, kinked tubing, loose connection
SpO2 no reading	Probe placement, perfusion, ambient light, probe LED failure, cable
Ventilator high-pressure alarm	Kinked circuit, secretions, patient coughing, water in tubing
Ventilator low-pressure or volume alarm	Disconnect, leak, cuff leak, circuit crack
Infusion pump occlusion alarm	Kinked tubing, closed clamp, clotted line, set loaded wrong
Infusion pump air-in-line alarm	Air in set, sensor dirty, set not seated
Defibrillator low delivered energy	Capacitor aging, relay, paddles or pads, battery
ESU patient burn	Return electrode contact, alternate path, damaged cable
Telemetry dropouts on one patient	Transmitter battery, leads, distance; many patients: antenna system or receiver, a system-wide problem
Monitor not sending to the central station or EMR	One bed: cable, port, network config on that monitor. Every bed: switch, server, middleware, network outage
Centrifuge vibration	Unbalanced load, worn rotor or bearing
Refrigerator temperature alarm	Door seal, defrost cycle, compressor, probe, overloading
EtCO2 no value or occlusion	Cannula and tubing, filter present, moisture trap (occlusion test), internal leak (occlude upstream of filter and trap); occlusion: kink, clogged filter, full trap
Respiration rate missing, ECG fine	Input printed circuit board; flat or short trace: raise sensitivity, then electrodes, leads, cable
Blood gas analyzer will not run	Cable and outlet first; 'too cold' after warming = thermistor; no battery run = charger; barcode dead = window
Centrifuge rotor will not spin	Lid not closed firmly (interlock), then connections, board, motor
Microtome sections wrong thickness	Blade clamp, dull edge, clearance angle; compressed = dull, warm block, too fast, wax
Tympanic thermometer reads low	Obstruction or cracked lens (high = torn cover); rest 30 minutes; ambient 10 to 40 C
Enteral pump low flow versus no set	Low flow: empty bag, occlusion, drip chamber. No set: seating, wrong set, bright light
PCA ignores requests	Settings (lockout, maximum) before pendant or motor; 'check settings' = door locked before setup
Aspirator suction fades	Leak (tubing, container, vacuum line) or clogging filter; sudden stop = fuse or overheated pump; none = disconnected, blocked filter, full container
ESU on but no output	High-voltage supply fuse

SYMPTOM	FIRST SUSPECTS
Tourniquet pump runs constantly	Leak in tubing, cuff, or valves, or a bad pressure transducer; will not inflate = pump disconnected
Autoclave no steam	Water pump, low fluid pressure, breaker or overload, burned-out heater, low water; over-pressure = pressure switch
Bilirubin light dark or dim	Too hot to relight; bulb then ballast; dirty lens; overheating = fan filter, fans, thermistor
Device went to battery while plugged in	Outlet and power cable continuity first, then the internal power circuit; regulator: measure input and output pins to ground

Priorities, communication, maintenance

- Priority order: life support and high-risk devices first, then patient risk, then urgency and downtime and clinical impact; loaners for critical downtime
- Device error versus use error: a device that performs to specification but was used wrongly needs education and possibly a design or labeling report, not a repair
- In-service and cross-training: teach proper use, care, cleaning, and alarm response; document attendance; communicate clearly with clinicians, staff, and manufacturers in writing and verbally
- Preventive maintenance: scheduled or risk-based; AEM programs allowed with documented justification (not imaging, lasers, or where the manufacturer requires); PM after repair; performance verification against tolerances
- Calibration traceable to national standards; test equipment has its own calibration schedule
- Incoming inspection before first use: physical, functional, safety test, inventory entry, manuals on file; lifecycle ends with data wiped, decontamination, and disposition
- Criticality = importance to the mission + likelihood of failure or PM demand + potential harm; repair time is not a factor; redundancy lowers priority but does not prevent all downtime; checklists guide user inspections
- Root cause analysis: what happened, why, how to prevent; keep asking why (five whys); ultrasound blurry = crystal then gel; oximeter erratic = ambient light; one sphygmomanometer = technique then cracked tubing (a leak reads low)
- Local versus network: one room missing = that room; several = shared configuration or network; false alarms on many monitors = alarm limits; check physical links first; search repair records for fleet patterns; ping, cable tester, multimeter help, a safety analyzer does not
- Switching supply shorted element = full input voltage out, open = zero; UPS short runtime = battery, no backup = inverter, switched over while plugged in = outlet or cable; desktop PSUs die of fan failure and heat

11 Healthcare information technology

NETWORKING FACT	VALUE
IPv4 address	Four numbers 0 to 255; subnet mask separates network and host; default gateway is the exit router; same subnet required to talk without a router
MAC address	48-bit hardware address in 12 hex digits, burned into the network card; switches forward by MAC
Private ranges	10.0.0.0/8; 172.16.0.0 to 172.31.255.255; 192.168.0.0/16
DHCP and static	DHCP assigns addresses automatically; medical devices often use static addresses or DHCP reservations so they are always reachable
DNS	Names to addresses; ping by address works but by name fails means DNS
APIPA	169.254.x.x means DHCP failed
Ports	HTTP 80, HTTPS 443, SSH 22, DICOM commonly 104 or 11112, HL7 commonly 2575 (site specific)
Devices	Switch connects devices in a LAN by MAC; router connects networks by IP; access point connects Wi-Fi clients; access point controller manages many APs; KVM switch shares one keyboard, video, and mouse among servers; virtual servers run many operating systems on one host; LAN is local, WAN spans sites, cloud is provider-hosted
Cabling	Cat 5e 1 Gbps, Cat 6A 10 Gbps, 100 m limit; fiber for longer runs; link light off means Layer 1; cable tracer finds the far end

NETWORKING FACT	VALUE
Wi-Fi	2.4 GHz longer range, channels 1, 6, 11; 5 GHz more channels, less interference; WPA2 or WPA3; enterprise mode uses 802.1X with RADIUS
Telemetry and RTLS	WMTS bands for patient telemetry; real-time location systems track equipment with RF, infrared, or ultrasound tags
OSI layers	1 physical (media, hubs), 2 data link (frames, MAC, switches), 3 network (IP, routers), 4 transport (TCP/UDP, ports), 5 session, 6 presentation (format conversion, encryption), 7 application (APIs); no conversion layer
Topologies	Bus and ring fail on any break; star (central switch) survives one bad cable; hybrid; mesh partial or full; point-to-multipoint (central node also processes); physical versus logical
Binary masks	255.255.255.240 = ...11110000 (/28, 16 addresses); 248 = 11111000; 252 = 11111100; ones must be contiguous; subnetting separates departments and limits broadcasts
Hub versus switch, PPP	Hub repeats to every port at layer 1, half duplex, collisions; switch keeps a source address table and forwards at layer 2, full duplex; PPP links two routers over serial or dial with authentication

COMMAND OR TOOL	USE
ping	Is the address reachable, and how fast; ping the gateway first
ipconfig (Windows), ifconfig or ip (Linux)	Show the device's address, mask, gateway; release and renew DHCP
tracert or traceroute	Every hop to a destination; where the path breaks
nslookup	Test name resolution
arp -a	IP to MAC table; duplicates
netstat	Open connections and listening ports
Cable tracer and tester	Find the far end; check pinout and continuity
TDR	Time-domain reflectometer: pulse reflection shows whether a cable is open and how far to the break
ping (ICMP)	Echo request and reply, IPv4 and IPv6; reports round-trip ms, packet loss, min, max, average, deviation; not a cable resistance test; power cycling is not a test
Fiber inspection microscope	Checks connector ends for dirt and damage; clean with isopropyl alcohol and lint-free wipes or cut and repolish; light source and power meter measure loss

INTEROPERABILITY	MEANING
HL7 version 2	Text messages between systems: ADT (admit, discharge, transfer), ORM (orders), ORU (results, including device data)
HL7 FHIR	Modern web-based standard using resources and APIs
DICOM	Standard for medical images and their transfer; modality worklist pulls patient data to the device; PACS stores and serves images
EMR and EHR	Electronic medical record (within one organization) and electronic health record (across); devices feed vitals in directly or through middleware
Middleware	Translates and routes between devices and the EMR; device gateways; where most integration faults are diagnosed
Time synchronization	NTP keeps device clocks aligned so records and alarms line up
PACS, RIS, LIS, LIMS	PACS stores and transports images with patient data (redundant servers); RIS schedules, tracks, and reports radiology; LIS tracks specimens and analyzer results; LIMS is the research and high-volume version

DICOM and HL7 detail	DICOM is a 22-part standard for image format and transfer referencing IEC definitions; HL7 standards: version 2 messaging, version 3 XML, Clinical Document Architecture; HL7 does not set leakage limits or breach rules
-----------------------------	---

Protected data and cybersecurity

- HIPAA: privacy rule (who may see PHI), security rule (administrative, physical, technical safeguards), breach notification; PHI is any identifiable health information; minimum necessary
- NIST Cybersecurity Framework functions: govern, identify, protect, detect, respond, recover
- HITRUST: a certifiable framework combining many requirements; HHS cybersecurity performance goals: essential and enhanced practices for hospitals (MFA, patching, backups, segmentation, training)
- MDS2: manufacturer disclosure statement for medical device security; SBOM: software bill of materials
- Controls: unique accounts and MFA, least privilege, encryption at rest and in transit, patching on the vendor's validated schedule, network segmentation for devices that cannot be patched, antivirus where the vendor allows, physical security of workstations, screen locks
- Backups: 3 copies, 2 media, 1 offsite; test restores; ransomware response: isolate, report, restore
- Incident: disconnect the device from the network, do not power off if forensics are needed, report to IT security and the vendor, document
- HITECH (2009): EHR incentives and penalties, meaningful use, breach notification to individuals and HHS, business associates bound, no sale of PHI, patient access; sets no spending guidelines
- HIPAA titles: I insurance portability; II administrative simplification with the privacy and security rules, fraud control, and fines; III medical savings accounts; IV enforcement by the Office for Civil Rights; V revenue offsets
- DMCA (1998): bans circumvention tools, raises penalties, limits provider liability; exemptions for security testing and personal implant data; does not ban copying devices as such
- FDA MDDS: systems that store, transfer, reformat, or display device data without controlling devices; reclassified Class III to Class I in 2011; a vital signs monitor yes, a lobby TV no
- IEC 80001-1 (2010, IEC and ISO): risk management for IT networks with medical devices; excludes drugs; applies to any organization's network; defines no acceptable risk level; not from the FDA
- MDS2 contents: automatic logoff, audit controls, authorization, configurable security, security updates, de-identification, backup and recovery, emergency (break-glass) access, data integrity, malware protection, node and user authentication, physical locks, third-party component plans, remote servicing; not structural integrity

Computer hardware and software troubleshooting

- Hardware: power supply (no power, random reboots), hard drive (slow, errors, clicking; SMART warnings; replace and restore image), memory (crashes), peripherals (drivers, cables, ports), connectors and cables (reseat, swap)
- Software: Windows, Office, Linux, and Unix updates and patches only on the vendor's validated schedule for medical devices; drivers after updates; event logs; restore from image; user accounts and permissions
- Order of checks: power and cables, link lights, address and gateway, ping the gateway, ping the server, name resolution, application or service, then escalate to IT or the vendor